Seeing the Small Bowel in Children: Capsule Endoscopy, Patency Testing and Balloon Enteroscopy.

Picture three children on the same clinic list. The first is a twelve-year-old boy with known Crohn's disease of the terminal ileum who now has cramping pain after meals, and whose parents ask whether he can simply swallow the camera they have read about. The second is a four-year-old girl with recurrent black stools, iron deficiency anemia and a normal upper endoscopy and colonoscopy, who cannot swallow a tablet. The third is a nine-year-old boy with Peutz-Jeghers syndrome, an inherited condition of intestinal polyps, whose magnetic resonance scan shows a two-centimeter polyp in the jejunum. Each child needs the small bowel examined, and each needs a different tool, used in a different order, for a different reason.

The clinical question of this chapter is how to choose and sequence the tools that see the small bowel in children: the video capsule, the patency capsule that tests whether a capsule can pass, endoscopic delivery of the capsule for the child who cannot swallow it, and balloon-assisted enteroscopy, which reaches deep into the small bowel with a scope that can take biopsies and treat. The main idea is this. The capsule is a diagnostic tool, and its main risk is retention, meaning the capsule gets stuck. That risk follows the disease, above all stricturing Crohn's disease, and not the age or size of the child. Balloon enteroscopy is the therapeutic partner of the capsule, not its competitor. So the practical sequence is to image first when obstruction is possible, to test patency in known Crohn's disease, to place the capsule endoscopically in the child who cannot swallow it, and to send anything that needs treatment to balloon enteroscopy.

We will move through the topic in order. First, how the tools physically reach the bowel and the limits that follow. Second, what the capsule finds in children. Third, retention: who is at risk, how the patency capsule works, and what to do when a capsule stays. Fourth, swallowing versus endoscopic placement. Fifth, balloon enteroscopy in children, its results and its complications, with Peutz-Jeghers syndrome as the leading example. We then put the pieces into a pathway, compare the guidelines, list the traps, and finish with what to do differently in clinic.

A note on sources. The pediatric guideline is the 2017 joint guideline on pediatric gastrointestinal endoscopy from the European Society of Gastrointestinal Endoscopy, known as ESGE, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, led by Tringali and Thomson. In North America, the 2017 capsule endoscopy clinical report of the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, was written by Friedlander and colleagues. For adults and mixed populations we use the 2023 update of the

ESGE small bowel guideline and the 2015 American College of Gastroenterology guideline on small bowel bleeding by Gerson and colleagues. For pediatric inflammatory bowel disease we use the revised Porto criteria of 2014 by Levine and colleagues. The core pediatric data come from the 2011 meta-analysis by Cohen and Klevens, the 2020 meta-analysis of retention in Crohn's disease by Pasha and colleagues, a large Chinese pediatric series by Li and colleagues in 2024, and pediatric balloon enteroscopy series from the United Kingdom, Japan and China.

Part one. How the tools reach the bowel.

The video capsule is a small camera with its own light and battery, either swallowed or placed into the duodenum with an endoscope. Once in the small bowel, it is carried along by the normal muscular waves of the gut, called peristalsis, and records the lining as it goes. No air is pumped in and no sedation is needed if the child swallows it. The capsule can see but cannot touch: it cannot take a biopsy and cannot treat anything.

The capsule's main hazard follows from its physical nature. A narrowing that lets liquid and soft food through may still stop a rigid capsule. When the capsule stays in the gut for two weeks or longer, or when it has to be removed sooner, we call that retention. The patency capsule exists to test for this in advance. It has the same size and shape as the video capsule but is made of a material that begins to dissolve about thirty hours after it is swallowed, over roughly twelve hours, according to the original 2006 clearance by the United States Food and Drug Administration, known as the FDA. It also contains a small radio tag so that it can be located. If it passes intact, or is found intact in the colon at around thirty hours, the bowel is patent enough for the video capsule. If it is held up, it dissolves rather than causing an obstruction.

Balloon enteroscopy works on a different principle. A long scope runs inside a flexible overtube, and one or two balloons alternately grip the bowel wall. By pushing and pulling, the endoscopist pleats the small bowel onto the scope, a little like gathering a curtain onto a rod, so that the scope can advance deep into the jejunum from the mouth or into the ileum from the anus. Because it is a real endoscope, it can take biopsies, remove polyps, dilate strictures, stop bleeding and retrieve a retained capsule. Size is the limitation. In a prospective Japanese multicenter series reported by Hagiwara and colleagues in 2019, the smallest children examined weighed ten point eight kilograms by the anal route and twelve point nine kilograms by the oral route. A 2025 review by Hoskins cites lower limits of about twelve kilograms for double-balloon and thirteen and a half kilograms for single-balloon enteroscopy. A third technique, spiral enteroscopy, has no pediatric data, and its sixteen-millimeter overtube is unsuitable for small children.

Part two. What the capsule finds.

The best summary of pediatric capsule endoscopy remains the 2011 meta-analysis by Cohen and Klevens, which pooled seven hundred and forty procedures in seven hundred and twenty-three patients aged eighteen or younger. About two thirds of studies had positive findings, sixty-five point four percent. Sixty-nine percent led to a new diagnosis and sixty-eight percent to a change in treatment. The capsule reached the colon, giving a complete small bowel examination, in eighty-six percent. Inflammatory bowel disease was the indication in fifty-four percent, about a third suspected Crohn's disease and a sixth known Crohn's disease, with obscure bleeding and polyposis making up most of the rest. Retention occurred in two point six percent, half a percent in the stomach and about two percent in the small bowel.

The yield depends on the question. In a European multicenter study of eighty-three children aged one and a half to almost eight years, reported by Fritscher-Ravens and colleagues in 2009, the slide showed five bars of similar height. The capsule was diagnostic in fifty-three percent of children with obscure bleeding, fifty-five percent with suspected Crohn's disease, fifty percent with abdominal pain, sixty-seven percent with protein-losing enteropathy, and fifty percent with malabsorption. None of these young children retained the capsule, and about three quarters had it placed endoscopically. A 2013 review by Cohen pooled pediatric yields by indication and found the highest yield in polyposis, about eighty percent, compared with about thirty-eight percent for obscure bleeding or iron deficiency anemia and forty-three percent for abdominal pain. Isolated abdominal pain without alarm features is a weak reason to swallow a camera.

In Crohn's disease, the capsule has two roles. The first is diagnosis and staging. The revised Porto criteria require small bowel imaging in every child with suspected inflammatory bowel disease, using either magnetic resonance enterography, usually shortened to MRE, or wireless capsule endoscopy, except in typical ulcerative colitis. The second role is monitoring. In a 2019 study by Oliva and colleagues, forty-eight children with Crohn's disease underwent pan-enteric capsule endoscopy, a capsule that examines both small bowel and colon, as part of a treat-to-target strategy. The slide drew this as three falling bars. Seventy-one percent had active inflammation at baseline, forty-six percent at week twenty-four and thirty-nine percent at week fifty-two, and treatment was changed after the baseline capsule in seventy-one percent. A useful way to think of the two modalities is that the capsule sees the surface, including superficial jejunal disease that MRE can miss, while MRE sees the wall, fistulas and strictures that the capsule cannot.

Part three. Retention, patency and what to do when the capsule stays.

The most important message about retention is that the risk follows the disease, not the age. The slide showed retention rates by indication as a ladder of bars. In children, retention was about one point two percent when the indication was polyps and one point four percent for

obscure bleeding. In pediatric Crohn's disease it was one point six four percent in the 2020 meta-analysis by Pasha and colleagues and about two point two percent in pooled pediatric data. In all ages, retention was two point three five percent in suspected Crohn's disease and four point six three percent in established Crohn's disease. In children with known inflammatory bowel disease in the 2009 series by Atay and colleagues it was five point two percent. Interestingly, children with Crohn's disease retained capsules less often than adults, one point six four percent compared with three point four nine percent, and in children, established and suspected Crohn's disease did not differ.

Some small groups had much higher risk. In the Atay data, as summarized in the 2013 review, retention was about thirty-six percent when a barium small bowel follow-through had already shown Crohn's disease, and forty-three percent in children with known inflammatory bowel disease and a body mass index below the fifth centile. In a 2012 pediatric series of two hundred and eighty-four procedures by Cohen and colleagues, seven capsules were retained and six of these children went to surgery. In contrast, among one thousand one hundred and forty Chinese children reported in 2024, only two retained the capsule, both with stenosing Crohn's disease, and both capsules were retrieved by enteroscopy. These numbers need careful reading. The very high rates come from small, selected groups and an earlier era; they tell us whom to image first rather than predicting an individual child's risk. And the management of retention has changed, from surgery toward medical treatment and enteroscopic retrieval.

That leads to the patency capsule. The 2023 ESGE update makes a strong recommendation to use a patency capsule before capsule endoscopy in established Crohn's disease, and in suspected Crohn's disease with obstructive symptoms. It also recommends dedicated cross-sectional imaging first when there are obstructive symptoms or a known stenosis, and does not recommend routine patency testing or imaging in suspected Crohn's disease without obstructive symptoms. The practical sequence is therefore: select the child at risk, image with MRE if a stenosis is likely, give the patency capsule, check at about thirty hours whether it has passed intact or reached the colon, and then proceed with the video capsule if the bowel is patent, or turn to enteroscopy or MRE if it is not.

The pediatric evidence is modest but reassuring. In a prospective study by Cohen and colleagues published in 2011, eighteen children aged ten to sixteen took the patency capsule. Fifteen passed it intact, at a mean of about thirty-four and a half hours, with a range of nine to sixty hours. Sixteen went on to a successful video capsule study with no retention, and every child whose patency capsule took forty hours or more had Crohn's disease. The patency capsule is not perfect, however. In established Crohn's disease, in all ages, retention was about two point nine percent after a patency capsule and about two point three percent after MR or CT enterography. Both strategies lower the risk; neither abolishes it.

When a capsule does stay, it is useful to remember that a retained capsule is also a finding. Most retained capsules cause no symptoms. The ESGE recommends conservative treatment first, which in Crohn's disease means treating the inflammation, and in one large series more than half of retained capsules passed this way. If the capsule does not pass, balloon enteroscopy is recommended to retrieve it, and the same procedure can dilate the stricture. Surgery is reserved for obstruction or failed retrieval, and then the culprit stricture is resected. The retained capsule has located the stricture, reclassified the disease as stricturing, and changed treatment. Consent before a capsule study should therefore name retention, its usual silence, and the routes to removal, endoscopic or rarely surgical.

Part four. Swallowing or placing the capsule.

Most children older than about eight can swallow the capsule. In the 2024 Chinese series, the slide showed three rising bars: twenty-nine percent of children aged five or younger swallowed the capsule, seventy-seven percent of those aged six to eleven, and eighty-four percent of those aged twelve or older. In a Japanese registry of over ten thousand capsule studies reported by Ohmiya and colleagues in 2022, the cut-offs for spontaneous swallowing in children under sixteen were an age of about nine years and two months, a height of one hundred and thirty-two centimeters and a weight of about twenty-five kilograms. In pooled pediatric studies, about eighty-eight percent of children swallowed the capsule, and the youngest reported child to do so was about four years old. Training helps. In a British study of pill-swallowing training published in 2021, a single session of about fifteen minutes taught eighty-seven percent of children to swallow pills, some as young as three, although this study was about medicines rather than capsules.

Below about eight, the capsule is usually placed. Endoscopic delivery uses a device that holds the capsule at the tip of the gastroscope and releases it into the duodenum. In a Japanese series reported by Oikawa-Kawamoto and colleagues in 2013, the youngest child was ten months old and weighed seven point nine kilograms, and the median age of children who needed placement was two years. Small bowel transit took longer after placement, about four hundred minutes compared with about two hundred and thirty when swallowed. In the large Japanese registry, delivery to the duodenum or jejunum succeeded in about ninety-two percent of children under sixteen, and mild adverse events, such as mucosal damage, minor bleeding or pain, occurred in about twenty-two percent of endoscopic deliveries. The reason to place rather than wait is that a capsule left in the stomach of a small child may not empty before the battery runs out. The trade-off is anesthesia and an upper endoscopy, which is best combined with the diagnostic gastroscopy the child often needs anyway.

Part five. Balloon enteroscopy in children.

The pediatric experience with balloon enteroscopy is small but consistent. In a prospective British series by Urs and colleagues in 2014, fifty-eight children underwent one hundred and thirteen double-balloon procedures at a median age of almost thirteen years. Polyposis and obscure bleeding were the main indications, the diagnostic yield was about seventy-one percent, and therapy succeeded in about forty-seven percent. There were three complications in fifty-eight children, about five percent, including one perforation in a child who had had a small bowel transplant. An earlier British series by Thomson and colleagues in 2010 examined fourteen children aged eight to seventeen, with a diagnostic yield of eleven of fourteen and no complications from the enteroscopy itself. The prospective Japanese multicenter series by Hagiwara and colleagues included seventy-nine children aged one to seventeen, with a diagnostic yield of forty-eight percent, two severe adverse events and no perforation. A Chinese series by Li and colleagues in 2020 focused on forty-one children with Peutz-Jeghers syndrome aged five to fourteen, with polypectomy in about eighty percent and a complication rate of about one percent per procedure.

The slide on complications drew these figures as bars, with a warning that the denominators differ, some per procedure and some per patient. The rates ranged from about one percent per procedure in the Chinese Peutz-Jeghers series and two and a half percent severe events per patient in the Japanese prospective series, to about five percent in the British series and in a large Japanese series of two hundred and fifty-seven procedures reported by Yokoyama and colleagues. In that Japanese series, the rate in children under ten was about ten percent. The complications that occur are pancreatitis, bleeding after polypectomy, perforation, often after previous surgery or transplant, and rarely bile duct injury. The risk is highest in children under ten, in altered anatomy, and with large polypectomies. This is work for high-volume centers with pediatric anesthesia and surgical back-up.

Peutz-Jeghers syndrome is the commonest therapeutic reason for balloon enteroscopy in children, and it shows how the tools fit together. The ESGE and ESPGHAN recommend starting small bowel surveillance, by capsule, MRE or both, no later than age eight, and earlier if there are symptoms. Polyps can then be removed electively by balloon enteroscopy. In the Chinese pediatric series, two hundred and forty-two polyps were removed from forty-one children, and ten children had giant polyps larger than five centimeters. For a heavy polyp burden, enteroscopy during laparotomy can be added; in mostly adult data cited by the ESGE, this raised clearance by sixteen percent. Polypectomy complication rates of about four to six percent are reported, and in children even polyps smaller than fifteen millimeters may cause trouble. The goal is elective clearance in the endoscopy suite instead of emergency laparotomy for intussusception, the telescoping of bowel around a polyp that brings these children to surgery.

Part six. Choosing the path.

Putting this together, a practical pathway has five steps. First, define the question: bleeding, Crohn's disease, polyps or protein loss, and what result would change management. Second, image with MRE first if obstruction or stricture is possible. Third, use a patency capsule in known Crohn's disease or with obstructive symptoms. Fourth, give the capsule, swallowed if the child is able, usually from about eight or nine years, and otherwise placed endoscopically. Fifth, send any lesion that needs biopsy, removal, dilation or hemostasis to balloon enteroscopy. Two shortcuts are worth knowing. Skip the capsule and go straight to enteroscopy when the target is already known, such as a polyp seen on MRE in Peutz-Jeghers syndrome or a stricture that needs dilation. And skip the enteroscope when the question is only extent or healing in Crohn's disease, because the capsule and MRE answer that without a long anesthetic.

Part seven. What the guidelines say.

On small bowel evaluation in suspected pediatric inflammatory bowel disease, North America and Europe are aligned: the NASPGHAN report recognizes the capsule for extent and recurrence, and the Porto criteria require MRE or capsule for all except typical ulcerative colitis. On suspected small bowel bleeding they are also aligned: the 2015 American College of Gastroenterology guideline in adults places the capsule after negative upper and lower endoscopy, and the 2017 ESGE and ESPGHAN guideline suggests capsule endoscopy in children with suspected small bowel bleeding and balloon enteroscopy for therapy. On patency testing before a capsule in known Crohn's disease they differ: the ESGE gives a strong recommendation for a patency capsule, while the North American documents reviewed contain no pediatric mandate. On the retained capsule they are aligned: the NASPGHAN report notes that retention is often silent and that surgery is occasionally required, and the ESGE recommends conservative care first and then enteroscopy. On minimum age or weight there is a material difference: the FDA cleared the capsule from age two in 2009, while no society sets a weight floor for pediatric balloon enteroscopy, so practice follows the experience of each center. Finally, on capsule endoscopy in acute upper gastrointestinal bleeding, the European guideline does not suggest routine use, and there is no North American pediatric statement, which is a gap.

Part eight. Traps, and what to do differently in clinic.

Four traps recur. The first is "too young": age is not the risk, the capsule can be placed, and it has been used from ten months and seven point nine kilograms. The second is "it is just a capsule": in stricturing Crohn's disease without MRE or a patency test, the result can be retention. The third is "the patency capsule passed": the risk is lowered, not zero, and a late or fragmented passage, or a capsule whose position was not confirmed, should be read with

caution. The fourth is "pain only": isolated abdominal pain without alarm features is a low-yield indication, and a camera will not fix it. And a negative capsule does not end the search when bleeding recurs; a repeat study, enteroscopy or MRE may still find the lesion.

To bring these lessons into clinic, consider five short scenarios. A twelve-year-old with known ileal Crohn's disease and pain after meals is at risk of a stricture: order MRE first, then a patency capsule before any video capsule. A four-year-old with recurrent melena and normal upper endoscopy and colonoscopy has suspected small bowel bleeding: place the capsule endoscopically during anesthesia, and plan enteroscopy for any lesion found. A nine-year-old with Peutz-Jeghers syndrome and a two-centimeter jejunal polyp on MRE needs elective clearance: go straight to balloon enteroscopy and polypectomy, and skip the capsule. A well child whose capsule is still in the ileum at two weeks has a retention that is also a finding: treat the inflammation, and retrieve the capsule by enteroscopy if it stays. And a seven-year-old with suspected Crohn's disease who cannot swallow tablets has a delivery problem: try pill-swallowing training, or place the capsule during the diagnostic endoscopy.

The whole chapter can be held in two sentences. The capsule is a safe, high-yield diagnostic tool in children of almost any age, as long as you image first and test patency when stricturing Crohn's disease is possible, and place it in the child who cannot swallow. When the answer needs a biopsy, a snare, a balloon dilator or a retrieval net, balloon enteroscopy in an experienced center is the next step, with complications that rise in the smallest children.

Questions to test yourself.

Question one. What did the 2011 pediatric meta-analysis show? Answer: in seven hundred and forty procedures, findings were positive in sixty-five percent, treatment changed in sixty-eight percent, the examination was complete in eighty-six percent, and retention occurred in two point six percent.

Question two. What drives the risk of capsule retention? Answer: the disease, above all stricturing Crohn's disease, not the child's age; pediatric Crohn's retention was about one point six percent, but it rose to about thirty-six percent when a barium study had already shown Crohn's disease.

Question three. How does the patency capsule work, and how is it read? Answer: it is the same size as the video capsule, begins to dissolve about thirty hours after swallowing, and passage intact or arrival in the colon at about thirty hours indicates that the small bowel is patent.

Question four. At what age do most children swallow the capsule? Answer: about three quarters of children aged six to eleven and most children over about nine years swallow it, while most children of five or younger need endoscopic placement.

Question five. How young can a capsule be placed endoscopically? Answer: it has been placed at ten months of age and seven point nine kilograms, although small bowel transit is slower after placement.

Question six. What are the results and risks of pediatric balloon enteroscopy? Answer: diagnostic yield of about fifty to seventy percent, complications of about one to five percent overall and about ten percent in children under ten, with pancreatitis, bleeding and perforation the main events.

Question seven. When should small bowel surveillance start in Peutz-Jeghers syndrome? Answer: no later than age eight, earlier if symptomatic, with elective enteroscopic polypectomy to prevent intussusception.

End of chapter.